# RELATORIO DE AVALIACAO E PREPARACAO - RENUMF90

Data de Processamento: 01/05/2026 as 06:18:10 ate 01/05/2026 as 06:18:24

## 1. CONFIGURACAO DO MODELO BIOLOGICO

Caracteristicas (Traits): PN_kg, PD_kg, PS_kg, GPD_g-dia, PE_cm, AOL_cm2, EGS_mm, MAR_%, CAR_kg-dia, PREC_SEX, IPP_dias, PROB_3P_%, STAY_%

Efeitos Fixos: GC, Fazenda, Estacao, Lote_Manejo, Regime_Alim, Tipo_Pastagem, Localidade_Bloco, Piquete, Sexo, Geracao, Raca

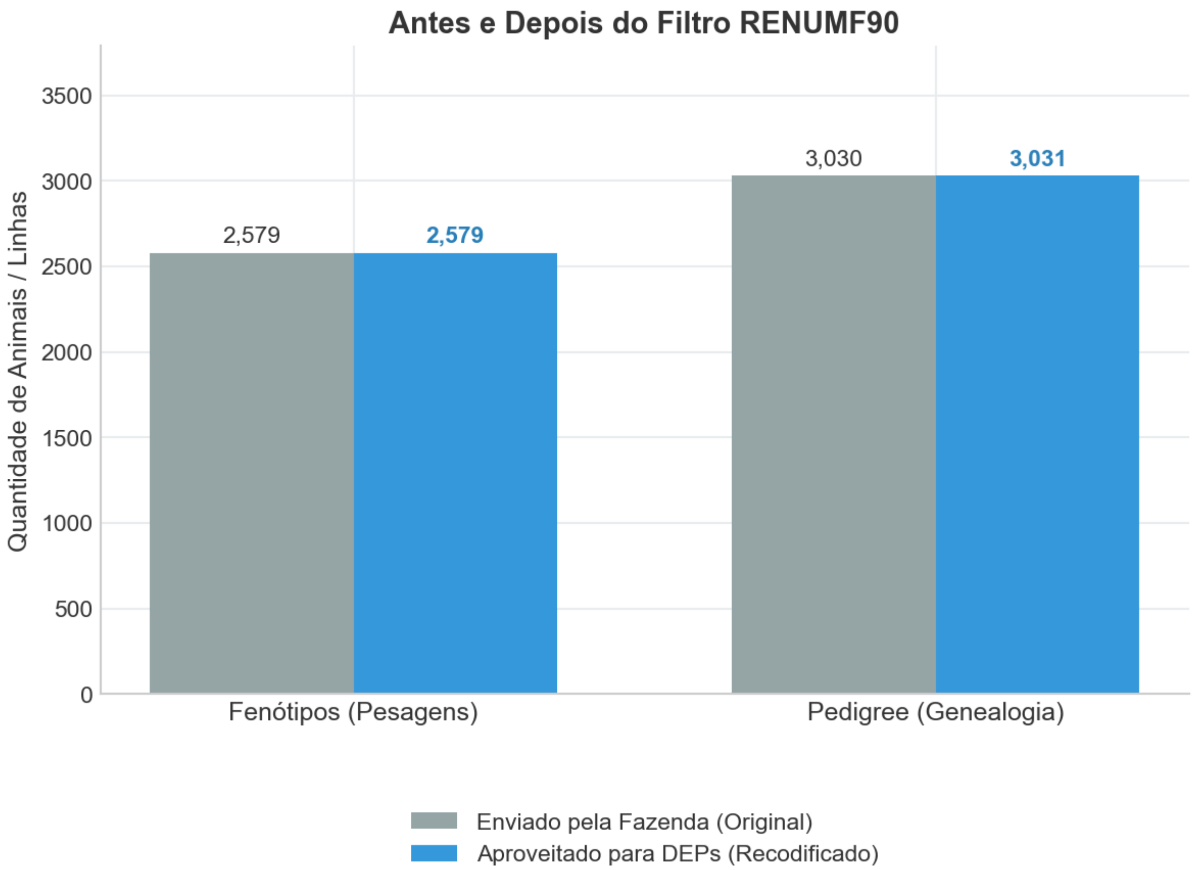
Covariaveis: itu_media, itu_dp, itu_max, itu_min, itu_leituras_criticas

## 2. RETENCAO E AUDITORIA DE DADOS

FENOTIPOS: Fornecidos (2579) -> Aproveitados (2579)

PEDIGREE: Fornecidos (3030) -> Aproveitados (3031)

GENOMICA: Total Analisado (3030) -> Isolados/Inuteis (0)




## 3. SAUDE GENETICA E ALERTAS

Consanguinidade Max: N/D | Media: N/D

# 4. LOG TECNICO ORIGINAL (MOTOR FORTRAN)

```
========================================
[PIPELINE LINEAR]
========================================
 RENUMF90 version 1.168
* START job     Wall time: 05-01-2026  06h 18m 11s 853
 name of parameter file? renum_linear.par
 datafile:fenotipos_raw.txt
 traits:           9          10          11          13          14
 R
   1.000       0.000       0.000       0.000       0.000
   0.000       1.000       0.000       0.000       0.000
   0.000       0.000       1.000       0.000       0.000
   0.000       0.000       0.000       1.000       0.000
   0.000       0.000       0.000       0.000       1.000

 Processing effect  1 of type cross
 item_kind=alpha

 Processing effect  2 of type cross
 item_kind=alpha

 Processing effect  3 of type cross
 item_kind=alpha

 Processing effect  4 of type cross
 item_kind=alpha

 Processing effect  5 of type cross
 item_kind=alpha

 Processing effect  6 of type cross
 item_kind=alpha

 Processing effect  7 of type cross
 item_kind=alpha

 Processing effect  8 of type cross
 item_kind=alpha

 Processing effect  9 of type cross
 item_kind=alpha

 Processing effect 10 of type cross
 item_kind=alpha

 Processing effect 11 of type cross
 item_kind=alpha

 Processing effect 12 of type cov

 Processing effect 13 of type cov

 Processing effect 14 of type cov

 Processing effect 15 of type cov

 Processing effect 16 of type cov

 Processing effect 17 of type cross
 item_kind=alpha
 pedigree file name  "pedigree_raw.txt"
 positions of animal, sire, dam, alternate dam, yob, and group     1     2     3     0     0     0     0
 SNP file name  "genotipos_raw.txt"
```

```
 Maximum size of character fields: 20

 Maximum size of record (max_string_readline): 8000

 Maximum number of fields for input file (max_field_readline): 100

 Pedigree search method (ped_search): convention

 Order of pedigree animals (animal_order): default

 Order of UPG (upg_order): default

 Missing observation code (missing): -999

  guessed number of lines         2579

 Using prime hash function
 hash tables for effects set up
 first 3 lines of the data file (up to 700 characters)
    1 122 2 2 3 3 2 3 4.2 3.5 -0.5 0 -999 28.21 0 1 1 1 4 1 1 20240322 2 114.3 75.37 6.82 87.31 -0.2056672425865524 133 1 1 1
20200101
    2 105 4 4 4 4 3 4 5.1 4.2 0.12 0 -999 72.98 0 1 1 2 1 2 1 20240226 38 114.3 77.56 5.64 86.99 1.5411790583507272 145 1 1 1
20200101
    3 141 1 1 1 2 1 1 3.8 3.1 -0.85 0 -999 83.38 0 1 2 3 1 3 1 20240611 41 114.3 68.62 5.72 78.72 -0.5160008312446829 72 1 1 1
20200101
% records read:      38.7747%, records read:       1000 timing           0.04687500
% records read:      77.5494%, records read:       2000 timing           0.09375000
 read         2579  records in   0.1093750      seconds
 ------------------------------
Processing effect group    1
 table with          108  elements sorted
 added count
 Effect group            1  of column            1  with          108  levels
 table expanded from        10000  to        10000  records
 Effect group            1  done in   0.6406250      seconds
 ------------------------------
Processing effect group    2
 table with            4  elements sorted
 added count
 Effect group            2  of column            1  with            4  levels
 table expanded from        10000  to        10000  records
 Effect group            2  done in   0.3437500      seconds
 ------------------------------
Processing effect group    3
 table with            5  elements sorted
 added count
 Effect group            3  of column            1  with            5  levels
 table expanded from        10000  to        10000  records
 Effect group            3  done in   0.3750000      seconds
 ------------------------------
Processing effect group    4
 table with           13  elements sorted
 added count
 Effect group            4  of column            1  with           13  levels
 table expanded from        10000  to        10000  records
 Effect group            4  done in   0.4375000      seconds
 ------------------------------
Processing effect group    5
 table with            4  elements sorted
 added count
 Effect group            5  of column            1  with            4  levels
 table expanded from        10000  to        10000  records
 Effect group            5  done in   0.3906250      seconds
 ------------------------------
Processing effect group    6
```

```
 table with            5  elements sorted
 added count
 Effect group            6  of column            1  with            5  levels
 table expanded from        10000  to        10000  records
 Effect group            6  done in   0.3593750      seconds
 ------------------------------
Processing effect group    7
 table with           10  elements sorted
 added count
 Effect group            7  of column            1  with           10  levels
 table expanded from        10000  to        10000  records
 Effect group            7  done in   0.5312500      seconds
 ------------------------------
Processing effect group    8
 table with          161  elements sorted
 added count
 Effect group            8  of column            1  with          161  levels
 table expanded from        10000  to        10000  records
 Effect group            8  done in   0.3906250      seconds
 ------------------------------
Processing effect group    9
 table with            2  elements sorted
 added count
 Effect group            9  of column            1  with            2  levels
 table expanded from        10000  to        10000  records
 Effect group            9  done in   0.3593750      seconds
 ------------------------------
Processing effect group   10
 table with            1  elements sorted
 added count
 Effect group           10  of column            1  with            1  levels
 table expanded from        10000  to        10000  records
 Effect group           10  done in   0.3437500      seconds
 ------------------------------
Processing effect group   11
 table with            1  elements sorted
 added count
 Effect group           11  of column            1  with            1  levels
 table expanded from        10000  to        10000  records
 Effect group           11  done in   0.3437500      seconds
 ------------------------------
Processing effect group   12
 ------------------------------
Processing effect group   13
 ------------------------------
Processing effect group   14
 ------------------------------
Processing effect group   15
 ------------------------------
Processing effect group   16
 ------------------------------
Processing effect group   17
 added count
 Effect group           17  of column            1  with         2579  levels
 Effect group           17  done in   0.0000000E+00  seconds
 wrote statistics in file "renf90.tables"
 reading data file for basic statistics
% records read:      38.7747%, records read:       1000 timing           0.04687500
% records read:      77.5494%, records read:       2000 timing           0.07812500

 Basic statistics for input data  (missing value code is '-999')
 Pos  Min         Max         Mean        SD                 N
   9    1.7600      6.8500      4.3522     0.74329        2579
  10    1.7500      5.1400      3.5173     0.48949        2579
  11   -1.0700     0.85000    -0.12120     0.33410        2579
  13    848.00      1072.0      988.57      56.053          30
```

```
  14    20.030      85.000      52.870      18.769        2579
  25   -999.00      77.960      57.430      114.76        2579
  26   -999.00      7.3200     -6.1178      107.74        2579
  27   -999.00      90.210      67.240      115.86        2579
  28   -999.00      1.8847     -11.611      107.14        2579
  29   -999.00      160.00      84.398      138.77        1990

 Correlation matrix
        9    10    11    13    14    25    26    27    28    29
  9   1.00  0.06 -0.01  0.21  0.01 -0.00 -0.00 -0.00 -0.00 -0.01
 10   0.06  1.00  0.02  0.14  0.03 -0.00 -0.00 -0.00 -0.00  0.00
 11  -0.01  0.02  1.00 -0.37 -0.00 -0.04 -0.04 -0.04 -0.04 -0.03
 13   0.21  0.14 -0.37  1.00  0.01 -0.07 -0.07 -0.07 -0.07 -0.31
 14   0.01  0.03 -0.00  0.01  1.00  0.01  0.01  0.01  0.01  0.00
 25  -0.00 -0.00 -0.04 -0.07  0.01  1.00  1.00  1.00  1.00  0.97
 26  -0.00 -0.00 -0.04 -0.07  0.01  1.00  1.00  1.00  1.00  0.97
 27  -0.00 -0.00 -0.04 -0.07  0.01  1.00  1.00  1.00  1.00  0.97
 28  -0.00 -0.00 -0.04 -0.07  0.01  1.00  1.00  1.00  1.00  0.97
 29  -0.01  0.00 -0.03 -0.31  0.00  0.97  0.97  0.97  0.97  1.00

 Counts of nonzero values (order as above)
       2579      2579      2579        30      2579      2579      2579      2579      2579      1990
       2579      2579      2579        30      2579      2579      2579      2579      2579      1990
       2579      2579      2579        30      2579      2579      2579      2579      2579      1990
         30        30        30        30        30        30        30        30        30        22
       2579      2579      2579        30      2579      2579      2579      2579      2579      1990
       2579      2579      2579        30      2579      2579      2579      2579      2579      1990
       2579      2579      2579        30      2579      2579      2579      2579      2579      1990
       2579      2579      2579        30      2579      2579      2579      2579      2579      1990
       2579      2579      2579        30      2579      2579      2579      2579      2579      1990
       1990      1990      1990        22      1990      1990      1990      1990      1990      1990
 basic_statistics done in   0.1093750      seconds

 random effect with SNPs 17
 type: animal
 file: genotipos_raw.txt
 read SNPs        3030  records
 Effect group           17  of column            1  with         3031  levels
 add_snp done in   4.6875000E-02  seconds
 reorder_animal done in   0.0000000E+00  seconds

 random effect  17
 type:animal
renumber_animal, start    Wall time: 05-01-2026  06h 18m 17s 591
 opened output pedigree file "renadd17.ped"
 first 3 lines of the file (up to 700 characters)
    1 3012 4055
    2 3005 4210
    3 3044 4102
 load_animals time   9.3750000E-02  seconds
 read         3030  pedigree records
 loaded            0  parent(s) in round           1
 load_animals time   4.6875000E-02  seconds
 read         3030  pedigree records

 Pedigree checks
compute_inbreeding time, start    Wall time: 05-01-2026  06h 18m 17s 736

 Computations for inbreeding coefficients
actual inbreeding computation, start    Wall time: 05-01-2026  06h 18m 17s 738
actual inbreeding computation, done     Wall time: 05-01-2026  06h 18m 17s 739
 Tiny negative value will be replaced with 0 considered as numerical error.
 Wrote inbreeding file "renf90.inb" with original id

 Inbreeding statistics:
 the maximum inbreeding coefficient    =  0.0000
```

```
 average inbreeding for inbred animals =     NaN   n = 0
                      for all animals =  0.0000   n = 3031
compute_inbreeding time, done     Wall time: 05-01-2026  06h 18m 17s 790
 loop writing renadd, time   7.8125000E-02  seconds

 Number of animals with records                  =         2579
 Number of animals with genotypes                =         3030
 Number of animals with records or genotypes     =         3031
 Number of animals with genotypes and no records =          452
 Number of parents without records or genotypes  =            0
 Total number of animals                         =         3031
renumber_animal, end      Wall time: 05-01-2026  06h 18m 17s 861

 Wrote cross reference IDs for SNP file "genotipos_raw.txt_XrefID"
 write_SNP_Xref done in   0.1093750      seconds
 write_map done in   0.0000000E+00  seconds

 Wrote parameter file "renf90.par"
 write renf90.par done in   0.0000000E+00  seconds
 writing renf90.dat
% records written:      38.7747%, records written:       1000 timing           0.14062500
% records written:      77.5494%, records written:       2000 timing           0.28125000
 Wrote renumbered data "renf90.dat" 2579 records
 write renf90.dat done in   0.3437500      seconds

 Number of records NOT present in pedigree file: 1
 Wrote field information "renf90.fields" for 22 fields in data
 save_renum_table done in   0.0000000E+00  seconds
* END  job     Wall time: 05-01-2026  06h 18m 18s 338


========================================
[PIPELINE CATEGÓRICO (LIMIAR)]
========================================
 RENUMF90 version 1.168
* START job     Wall time: 05-01-2026  06h 18m 18s 399
 name of parameter file? renum_categ.par
 datafile:fenotipos_raw.txt
 traits:           3           4           5           6           7           8
          12          15
 R
   1.000       0.000       0.000       0.000       0.000       0.000       0.000
   0.000
   0.000       1.000       0.000       0.000       0.000       0.000       0.000
   0.000
   0.000       0.000       1.000       0.000       0.000       0.000       0.000
   0.000
   0.000       0.000       0.000       1.000       0.000       0.000       0.000
   0.000
   0.000       0.000       0.000       0.000       1.000       0.000       0.000
   0.000
   0.000       0.000       0.000       0.000       0.000       1.000       0.000
   0.000
   0.000       0.000       0.000       0.000       0.000       0.000       1.000
   0.000
   0.000       0.000       0.000       0.000       0.000       0.000       0.000
   1.000

 Processing effect  1 of type cross
 item_kind=alpha

 Processing effect  2 of type cross
 item_kind=alpha

 Processing effect  3 of type cross
 item_kind=alpha
```

```
 Processing effect  4 of type cross
 item_kind=alpha

 Processing effect  5 of type cross
 item_kind=alpha

 Processing effect  6 of type cross
 item_kind=alpha

 Processing effect  7 of type cross
 item_kind=alpha

 Processing effect  8 of type cross
 item_kind=alpha

 Processing effect  9 of type cross
 item_kind=alpha

 Processing effect 10 of type cross
 item_kind=alpha

 Processing effect 11 of type cross
 item_kind=alpha

 Processing effect 12 of type cov

 Processing effect 13 of type cov

 Processing effect 14 of type cov

 Processing effect 15 of type cov

 Processing effect 16 of type cov

 Processing effect 17 of type cross
 item_kind=alpha
 pedigree file name  "pedigree_raw.txt"
 positions of animal, sire, dam, alternate dam, yob, and group     1     2     3     0     0     0     0
 SNP file name  "genotipos_raw.txt"

 Maximum size of character fields: 20

 Maximum size of record (max_string_readline): 8000

 Maximum number of fields for input file (max_field_readline): 100

 Pedigree search method (ped_search): convention

 Order of pedigree animals (animal_order): default

 Order of UPG (upg_order): default

 Missing observation code (missing): -999

  guessed number of lines         2579

 Using prime hash function
 hash tables for effects set up
 first 3 lines of the data file (up to 700 characters)
    1 122 2 2 3 3 2 3 4.2 3.5 -0.5 0 -999 28.21 0 1 1 1 4 1 1 20240322 2 114.3 75.37 6.82 87.31 -0.2056672425865524 133 1 1 1
20200101
    2 105 4 4 4 4 3 4 5.1 4.2 0.12 0 -999 72.98 0 1 1 2 1 2 1 20240226 38 114.3 77.56 5.64 86.99 1.5411790583507272 145 1 1 1
20200101
    3 141 1 1 1 2 1 1 3.8 3.1 -0.85 0 -999 83.38 0 1 2 3 1 3 1 20240611 41 114.3 68.62 5.72 78.72 -0.5160008312446829 72 1 1 1
20200101
```

```
% records read:      38.7747%, records read:       1000 timing           0.04687500
% records read:      77.5494%, records read:       2000 timing           0.09375000
 read         2579  records in   0.1250000      seconds
 ------------------------------
Processing effect group    1
 table with          108  elements sorted
 added count
 Effect group            1  of column           1  with          108  levels
 table expanded from        10000  to        10000  records
 Effect group            1  done in   0.4062500      seconds
 ------------------------------
Processing effect group    2
 table with            4  elements sorted
 added count
 Effect group            2  of column           1  with            4  levels
 table expanded from        10000  to        10000  records
 Effect group            2  done in   0.3906250      seconds
 ------------------------------
Processing effect group    3
 table with            5  elements sorted
 added count
 Effect group            3  of column           1  with            5  levels
 table expanded from        10000  to        10000  records
 Effect group            3  done in   0.5156250      seconds
 ------------------------------
Processing effect group    4
 table with           13  elements sorted
 added count
 Effect group            4  of column           1  with           13  levels
 table expanded from        10000  to        10000  records
 Effect group            4  done in   0.3906250      seconds
 ------------------------------
Processing effect group    5
 table with            4  elements sorted
 added count
 Effect group            5  of column           1  with            4  levels
 table expanded from        10000  to        10000  records
 Effect group            5  done in   0.3593750      seconds
 ------------------------------
Processing effect group    6
 table with            5  elements sorted
 added count
 Effect group            6  of column           1  with            5  levels
 table expanded from        10000  to        10000  records
 Effect group            6  done in   0.3906250      seconds
 ------------------------------
Processing effect group    7
 table with           10  elements sorted
 added count
 Effect group            7  of column           1  with           10  levels
 table expanded from        10000  to        10000  records
 Effect group            7  done in   0.3906250      seconds
 ------------------------------
Processing effect group    8
 table with          161  elements sorted
 added count
 Effect group            8  of column           1  with          161  levels
 table expanded from        10000  to        10000  records
 Effect group            8  done in   0.3906250      seconds
 ------------------------------
Processing effect group    9
 table with            2  elements sorted
 added count
 Effect group            9  of column           1  with            2  levels
 table expanded from        10000  to        10000  records
 Effect group            9  done in   0.5000000      seconds
```

```
 -----------------------------
Processing effect group   10
 table with           1  elements sorted
 added count
 Effect group           10  of column           1  with           1  levels
 table expanded from        10000  to        10000  records
 Effect group           10  done in   0.3906250      seconds
 -----------------------------
Processing effect group   11
 table with           1  elements sorted
 added count
 Effect group           11  of column           1  with           1  levels
 table expanded from        10000  to        10000  records
 Effect group           11  done in   0.3593750      seconds
 -----------------------------
Processing effect group   12
 -----------------------------
Processing effect group   13
 -----------------------------
Processing effect group   14
 -----------------------------
Processing effect group   15
 -----------------------------
Processing effect group   16
 -----------------------------
Processing effect group   17
 added count
 Effect group           17  of column           1  with        2579  levels
 Effect group           17  done in   0.0000000E+00  seconds
 wrote statistics in file "renf90.tables"
 reading data file for basic statistics
% records read:      38.7747%, records read:       1000 timing           0.06250000
% records read:      77.5494%, records read:       2000 timing           0.10937500

 Basic statistics for input data  (missing value code is '-999')
 Pos  Min        Max        Mean       SD                N
   3    1.0000      4.0000      2.2687      1.0527        2579
   4    1.0000      4.0000      2.4781      1.1199        2579
   5    1.0000      4.0000      2.4897      1.1163        2579
   6    1.0000      4.0000      2.4843      1.1152        2579
   7    1.0000      4.0000      2.2762      1.0243        2549
   8    1.0000      4.0000      2.4986      1.1191        2579
  12    0.0000      1.0000     0.58162E-02 0.76057E-01    2579
  15    0.0000      1.0000     0.77549E-02 0.87737E-01    2579
  25   -999.00      77.960      57.430      114.76        2579
  26   -999.00      7.3200     -6.1178      107.74        2579
  27   -999.00      90.210      67.240      115.86        2579
  28   -999.00      1.8847     -11.611      107.14        2579
  29   -999.00      160.00      84.398      138.77        1990

 Correlation matrix
         3     4     5     6     7     8    12    15    25    26    27    28    29
   3   1.00  0.16  0.16  0.10  0.22  0.13 -0.02 -0.03  0.03  0.03  0.03  0.03  0.03
   4   0.16  1.00  0.16 -0.16  0.23  0.14 -0.03 -0.05  0.05  0.05  0.05  0.05  0.05
   5   0.16  0.16  1.00  0.87  0.12  0.12  0.00 -0.00 -0.05 -0.05 -0.05 -0.05 -0.06
   6   0.10 -0.16  0.87  1.00  0.04  0.06 -0.04 -0.05  0.05  0.05  0.05  0.05  0.05
   7   0.22  0.23  0.12  0.04  1.00  0.14   NaN   NaN  0.07  0.07  0.06  0.07  0.07
   8   0.13  0.14  0.12  0.06  0.14  1.00  0.00 -0.00 -0.05 -0.05 -0.05 -0.05 -0.05
  12  -0.02 -0.03  0.00 -0.04   NaN  0.00  1.00  0.63 -0.47 -0.47 -0.47 -0.47 -0.53
  15  -0.03 -0.05 -0.00 -0.05   NaN -0.00  0.63  1.00 -0.57 -0.57 -0.57 -0.57 -0.62
  25   0.03  0.05 -0.05  0.05  0.07 -0.05 -0.47 -0.57  1.00  1.00  1.00  1.00  0.97
  26   0.03  0.05 -0.05  0.05  0.07 -0.05 -0.47 -0.57  1.00  1.00  1.00  1.00  0.97
  27   0.03  0.05 -0.05  0.05  0.06 -0.05 -0.47 -0.57  1.00  1.00  1.00  1.00  0.97
  28   0.03  0.05 -0.05  0.05  0.07 -0.05 -0.47 -0.57  1.00  1.00  1.00  1.00  0.97
  29   0.03  0.05 -0.06  0.05  0.07 -0.05 -0.53 -0.62  0.97  0.97  0.97  0.97  1.00
```

```
 Counts of nonzero values (order as above)
       2579      2579      2579      2579      2549      2579      2579      2579      2579      2579      2579      2579
1990
       2579      2579      2579      2579      2549      2579      2579      2579      2579      2579      2579      2579
1990
       2579      2579      2579      2579      2549      2579      2579      2579      2579      2579      2579      2579
1990
       2579      2579      2579      2579      2549      2579      2579      2579      2579      2579      2579      2579
1990
       2549      2549      2549      2549      2549      2549      2549      2549      2549      2549      2549      2549
1968
       2579      2579      2579      2579      2549      2579      2579      2579      2579      2579      2579      2579
1990
       2579      2579      2579      2579      2549      2579      2579      2579      2579      2579      2579      2579
1990
       2579      2579      2579      2579      2549      2579      2579      2579      2579      2579      2579      2579
1990
       2579      2579      2579      2579      2549      2579      2579      2579      2579      2579      2579      2579
1990
       2579      2579      2579      2579      2549      2579      2579      2579      2579      2579      2579      2579
1990
       2579      2579      2579      2579      2549      2579      2579      2579      2579      2579      2579      2579
1990
       2579      2579      2579      2579      2549      2579      2579      2579      2579      2579      2579      2579
1990
       1990      1990      1990      1990      1968      1990      1990      1990      1990      1990      1990      1990
1990
 basic_statistics done in   0.1250000      seconds

 random effect with SNPs 17
 type: animal
 file: genotipos_raw.txt
 read SNPs        3030  records
 Effect group           17  of column            1  with         3031  levels
 add_snp done in   4.6875000E-02  seconds
 reorder_animal done in   0.0000000E+00  seconds

 random effect  17
 type:animal
renumber_animal, start    Wall time: 05-01-2026  06h 18m 24s 165
 opened output pedigree file "renadd17.ped"
 first 3 lines of the file (up to 700 characters)
    1 3012 4055
    2 3005 4210
    3 3044 4102
 load_animals time   9.3750000E-02  seconds
 read         3030  pedigree records
 loaded            0  parent(s) in round            1
 load_animals time   4.6875000E-02  seconds
 read         3030  pedigree records

 Pedigree checks
compute_inbreeding time, start    Wall time: 05-01-2026  06h 18m 24s 303

 Computations for inbreeding coefficients
actual inbreeding computation, start    Wall time: 05-01-2026  06h 18m 24s 305
actual inbreeding computation, done     Wall time: 05-01-2026  06h 18m 24s 306
 Tiny negative value will be replaced with 0 considered as numerical error.
 Wrote inbreeding file "renf90.inb" with original id

 Inbreeding statistics:
 the maximum inbreeding coefficient    =  0.0000
 average inbreeding for inbred animals =     NaN   n = 0
                       for all animals =  0.0000   n = 3031
compute_inbreeding time, done     Wall time: 05-01-2026  06h 18m 24s 369
 loop writing renadd, time   6.2500000E-02  seconds
```

```
 Number of animals with records                  =         2579
 Number of animals with genotypes                =         3030
 Number of animals with records or genotypes     =         3031
 Number of animals with genotypes and no records =          452
 Number of parents without records or genotypes  =            0
 Total number of animals                         =         3031
renumber_animal, end      Wall time: 05-01-2026  06h 18m 24s 438

 Wrote cross reference IDs for SNP file "genotipos_raw.txt_XrefID"
 write_SNP_Xref done in   0.1093750      seconds
 write_map done in   0.0000000E+00  seconds

 Wrote parameter file "renf90.par"
 write renf90.par done in   1.5625000E-02  seconds
 writing renf90.dat
% records written:      38.7747%, records written:       1000 timing           0.09375000
% records written:      77.5494%, records written:       2000 timing           0.17187500
 Wrote renumbered data "renf90.dat" 2579 records
 write renf90.dat done in   0.2187500      seconds

 Number of records NOT present in pedigree file: 1
 Wrote field information "renf90.fields" for 25 fields in data
 save_renum_table done in   0.0000000E+00  seconds
* END  job     Wall time: 05-01-2026  06h 18m 24s 782
```